REMEMBERING WHERE TO LOOK: CONTINUAL HIERARCHICAL EVIDENCE SELECTION FOR WHOLE-SLIDE IMAGES

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ABSTRACT

When diagnostic tasks arrive sequentially, does a whole-slide-image (WSI) model forget *where to look*? We isolate this question with an austere design: frozen CONCH encoders, a frozen prompt-based diagnosis head, no trained classifier—every performance change is attributable to one trainable object, a hierarchical evidence-selection policy (two small MLPs, 0.79M parameters). Continual learning touches only this policy: each task trains a rank-4 LoRA residual ($\approx 2.1\%$ of parameters) folded exactly into one shared selector at the task boundary, giving task-ID-free, constant-cost inference. Merging alone is only a substrate—without preservation losses it shows no systematic preservation benefit over sequential fine-tuning (3/5 paired seeds). Preservation comes from replay, dual-level selection distillation, and a utility hinge over stored behavioral snapshots (scores and indices; no images). On four TCGA tasks the full method gains $+34.6$ class-IL points over sequential fine-tuning (5/5 seeds), halves forgetting versus replay alone, and retains substantially more of its originally selected evidence than unprotected baselines (Jaccard 0.16 vs. 0.001).

***Index Terms*—** Whole-slide images, continual learning, evidence selection, low-rank adaptation, computational pathology

1. INTRODUCTION

A single WSI contains 10^4–10^5 patches, of which a diagnosis typically depends on a handful. Systems therefore acquire evidence under a budget: choose B patches, then decide. In deployment, tasks arrive sequentially—a hospital adds esophageal cases this quarter, renal the next—and retraining from scratch on all past cohorts is rarely feasible. Continual learning (CL) for WSIs has so far inherited its framing from natural-image classification: preserve the *classifier* or the *representation*. This framing obscures a failure mode unique to budgeted pathology pipelines: a model whose classifier is intact can still stop looking at the right tissue.

We make that failure mode measurable, and then substantially mitigate it. The design principle is **attribution by freezing**: the CONCH image and text encoders [1] are frozen; the diagnosis head is a fixed cosine rule against class-text embeddings; there is no trained classifier. The only trainable object is a hierarchical evidence-selection policy. Consequently, when methods are compared on the same data and label space, differences in accuracy, leakage, and evidence choice are attributable to the selection policy alone; temporal class-incremental changes additionally reflect the expanding label space, which is why we also measure selection drift directly.

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Prior LoRA-based CL—Online-LoRA [2], InfLoRA [3], SD-LoRA [4], CL-LoRA [5], FlexEM [6]—preserves classifier or backbone representations. We instead confine continual learning to the evidence-selection policy over a frozen diagnostic pathway, which lets us measure directly whether a model forgets *where to look for evidence*.

Contributions. (1) A formulation of continual evidence selection with complete attribution: frozen backbone, frozen head, selection-only training, and selection-level forgetting metrics (selection Jaccard, utility retention) alongside standard accuracy and forgetting. (2) A task-free consolidated selector: per-task rank-4 LoRA residuals folded exactly into one shared policy at task boundaries; inference uses a single model, no task ID, no memory, no module retrieval, at constant cost. (3) A mechanism separation supported by controls: merging is a substrate, not a demonstrated defense—LoRA-and-merge alone shows no systematic preservation benefit over sequential fine-tuning (3/5 paired seeds)—while preservation comes from replay, dual-level selection distillation, and a utility hinge operating on stored *behavioral snapshots* rather than stored images. (4) An evaluation with pre-registered gates and win-count statistics over five seeds, including honest reporting of removed components, of comparisons that did not reach systematicity, and of one pre-registered expectation our data refuted.

2. RELATED WORK

Evidence selection in WSIs. Multiple-instance learning aggregates patch features with learned attention [7, 8, 9], implicitly ranking evidence but entangling selection with a trained classifier. We make selection explicit and budgeted, and remove the trained classifier entirely so that selection is the sole learned behavior.

LoRA-based continual learning. LoRA [10] has become the vehicle of choice for parameter-efficient CL. Online-LoRA [2] expands adapters on detected distribution shift; FlexEM [6] maintains a module pool with dual-stage retrieval at inference; InfLoRA [3] and O-LoRA [11] constrain per-task subspaces; CL-LoRA [5] splits task-shared and task-specific adapters; SD-LoRA [4] decouples magnitude and direction and, like us, deploys the final model directly without per-task component selection. These methods differ in *when* adaptation is consolidated. Pool-and-retrieve methods keep modules separate and select at inference; deferred-composition methods combine independently trained task vectors post hoc [12, 13, 14]; we consolidate *incrementally*: each residual is trained on top of the already-merged selector under preservation constraints, then folded in exactly. Our study object also differs: prior work preserves classifier or backbone representations, whereas we preserve a selection *policy* over a frozen pathway.

Classical CL families. We compare against representatives of rehearsal [15, 16, 17] and functional regularization [18] adapted to

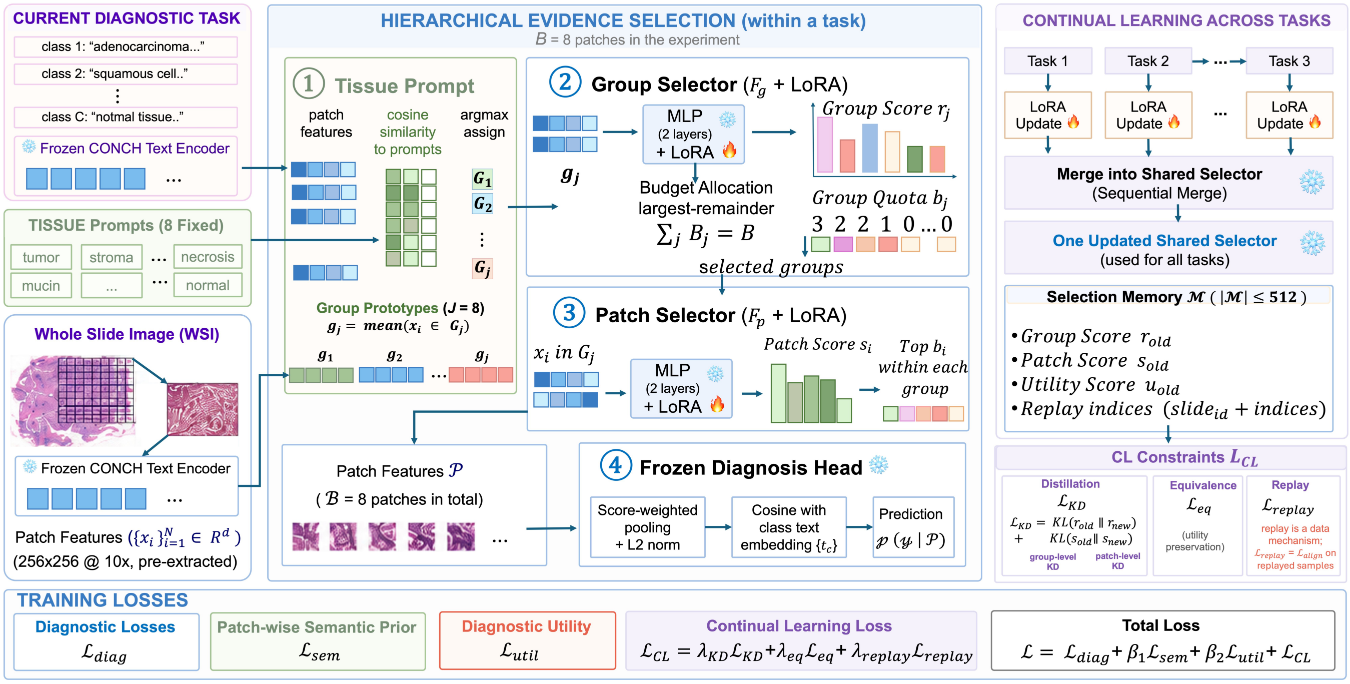

Fig. 1. **Continual hierarchical evidence selection.** *Left, middle (within a task):* patches from a frozen CONCH image encoder are assigned to J=8 tissue groups by argmax cosine similarity to fixed tissue prompts; a group selector F_g scores groups, a largest-remainder allocation converts scores into integer quotas $\sum_j b_j = B$ in one shot, and a patch selector F_p picks the top b_j patches within each group. Selected evidence is pooled (score-weighted, ℓ_2-normalized) and classified by a *frozen* cosine rule against class-text embeddings—no trained classifier exists anywhere in the pipeline. *Right (across tasks):* each task trains only rank-4 LoRA adapters on F_g, F_p (fire); at the boundary the residual is folded exactly into the shared selector, which is the only model used at inference (no task ID, no memory, no retrieval). A 512-entry Selection Memory stores behavioral snapshots—group/patch scores, utilities, and replay indices; *no patches or features*—and drives selection distillation, utility preservation, and replayed diagnostic training. Base MLPs are frozen within a task and updated *only* by the boundary merge.

the selection setting; parameter-importance methods [19] are complementary and left to future work (Sec. 8).

3. METHOD

3.1. Problem setup and the attribution principle

Tasks $\tau = 1, \ldots, T$ arrive sequentially; task τ contributes WSIs with labels from its own class set (two classes per organ task in our benchmark). For a slide with patch features $\{x_i\}_{i=1}^N \subset \mathbb{R}^{512}$ (frozen CONCH image encoder, 256×256 patches at $10\times$, pre-extracted), the system must select a set $\mathcal{P}$ with $|\mathcal{P}| = B = 8$ and predict from $\mathcal{P}$ alone. Prediction is a fixed rule: score-weighted pooling of $\mathcal{P}$, ℓ_2 normalization, cosine similarity to frozen class-text embeddings $\{t_c\}$, softmax over the classes under evaluation. **Training is task-aware; inference is task-free**: at test time the model receives no task identity and is evaluated class-incrementally over the union of all classes seen so far, as well as task-incrementally within each task's class pair. Because all non-selector parameters are frozen, cross-method differences evaluated on the same data and label space are attributable to selector updates; temporal class-incremental changes additionally reflect the expanding evaluation label space. We therefore also report *leakage*, the fraction of predictions falling in the wrong task's class set, and complement it with direct selection-drift metrics that bypass the head entirely.

3.2. Hierarchical selection within a task

Patches are grouped by semantics, not clustering: each x_i is assigned to $\arg\max_j \cos(x_i, p_j)$ over J=8 fixed tissue prompts (tumor, stroma, necrosis, ..., normal) encoded once by the frozen text encoder; group prototypes are means, $g_j = \mathrm{mean}(x_i \in G_j)$. Two structurally identical two-layer MLPs, $\mathrm{Linear}(1537{\to}256) \to \mathrm{GELU} \to \mathrm{Linear}(256{\to}1)$, implement the policy: the group selector F_g maps g_j to a score r_j; a largest-remainder (Hamilton) allocation converts $\mathrm{softmax}(r)$ into integer quotas b_j with $\sum_j b_j = B$ in a single shot; the patch selector F_p scores patches, and the top b_j within each group are selected. Each selector operates on its active 512-d visual input (g_j for F_g, x_i for F_p), with fixed zero padding retained only for implementation compatibility with the module's original 1537-d input shape; the discarded task-query and state-conditioning variants behind that shape are described in Sec. 7. The two MLPs total 787,970 parameters.

Within-task training minimizes $\mathcal{L}_{\mathrm{ev}} = \mathcal{L}_{\mathrm{diag}} + \beta_s\mathcal{L}_{\mathrm{sem}} + \beta_u\mathcal{L}_{\mathrm{util}}$: the frozen-head diagnostic loss on the selected evidence; a patch-wise semantic prior aligning patch scores with prompt affinities; and a differentiable utility term rewarding patches with positive counterfactual gain. Selection is made differentiable through the score-weighted pooling; $\beta_s = \beta_u = 0.1$ throughout.

3.3. Continual protocol: exact incremental consolidation

Per-task residuals, one shared selector. For every task, rank-4 LoRA adapters [10] are attached to each linear layer of F_g and F_p (four layers; 16,400 trainable parameters per task, $\approx 2.1\%$ of the selector; $\alpha = r$). During task τ the base is frozen and only the adapters train; the forward pass materializes $W_{\text{eff}} = W + BA$ at every step, so folding the residual in at the boundary,

$$W_\tau \;=\; W_{\tau-1} + \Delta W_\tau, \qquad \Delta W_\tau = B_\tau A_\tau, \tag{1}$$

is an *identity rewrite*: the network computes the same function immediately before and after the merge. The adapter is then re-initialized for the next task. Inference uses the single merged selector: no task ID, no adapter bank, no retrieval, constant parameter count.

Task-1 protocol, stated explicitly. In the main configuration the very first task also trains only LoRA, on a frozen *randomly initialized* base. Two observations contextualize this choice without settling it: the output layer's residual is a 1×256 matrix of rank at most one, hence unconstrained by the rank-4 budget; and A2 reaches final task-IL parity with full-parameter A1 (0.8105 vs. 0.8086; Table 1)—though this compares whole four-task lifecycles and does not isolate the task-1 initialization. Whether warm-starting the base is preferable is an empirical question; we evaluate it directly with the pre-registered warm-start ablation (task 1 trains the full base, LoRA thereafter) under a two-sided decision rule; the ablation found no systematic class-IL difference (warm start better in 2/5 seeds; Sec. 6), so the parameter-efficient LoRA-from-task-1 protocol is retained.

Merging is a substrate, not a mechanism. LoRA merging alone does not prevent forgetting: with adapters and boundary merges but no preservation losses, final class-incremental accuracy shows no systematic difference from naive sequential fine-tuning (0.4466 vs. 0.4774; 3/5 seeds, wide seed spread). What merging buys is a fixed-cost sequential *substrate*—a single consolidated policy with task-free inference. Preservation comes from the training-time losses of Sec. 3.4. Two boundary-related equivalences preempt confusion. First, "merge at each boundary" and "keep earlier adapters frozen in place and merge once at the end" compute the same function and, with matched initialization, the same training trajectory; we verify this to numerical tolerance with a unit test and spend no experiments on it. Second, composing *independently trained* task residuals post hoc (task arithmetic [12]) is *not* the same algorithm: each residual then never conditions on prior knowledge, and the composed model never receives a training signal. We evaluate this as a separate family in Sec. 6.

3.4. Selection Memory and preservation losses

The memory $\mathcal{M}$ ($|\mathcal{M}| \leq 512$, global reservoir sampling—uniform over all stored experiences) holds *behavioral snapshots*: per slide, the task tag, slide ID, terminal selection state, group scores r_{old}, candidate indices, patch scores s_{old}, and counterfactual utilities u_{old}. *No patches and no features are stored*; features are reloaded by slide ID when a snapshot is replayed. Because entries are sampled uniformly over the stream, each past task's share is proportional to what it contributed; capacity is deliberately small, and Sec. 5 shows the mechanism matters more than the capacity.

During task τ, each optimizer step draws one stored slide (replay_k=1, i.e., a 1:1 old:new ratio) and adds

$$\mathcal{L}_{CL} = \lambda_{kd}\big[\mathrm{KL}(r_{\text{old}} \| r_{\text{new}}) + \mathrm{KL}(s_{\text{old}} \| s_{\text{new}})\big] + \lambda_{eq}\max(0, U_{\text{old}} - U_{\text{new}}) + \lambda_r \mathcal{L}^{\text{rep}}_{\text{diag}}, \tag{2}$$

with $\lambda_{kd} = \lambda_{eq} = \lambda_r = 1$. Three design points deserve emphasis. The distillation teacher is *not a second network*: targets are the stored score snapshots, so no teacher forward pass or checkpoint is kept. The hinge penalizes only utility *regression*, leaving improvement unpenalized. And the replay term is the ordinary task loss evaluated on replayed slides—replay is a data mechanism, not a new objective; all three terms join the current sample's loss in a *single* backward pass and optimizer step, so replay arms take no extra gradient steps. With replay, distillation, and utility preservation disabled, the continual pipeline reduces exactly to the A2 LoRA-and-boundary-merge baseline. At inference the memory is not consulted at all.

4. EXPERIMENTAL SETUP

Benchmark. Four TCGA organ tasks—ESCA, RCC, BRCA, LUNG—each a binary subtype problem; the primary sequence is ESCA→RCC→BRCA→LUNG. Train/test: 2,273 slides train; test $n = 15/76/93/95$ per task. Class-incremental evaluation uses the union of all classes seen so far; task-incremental uses each task's own pair; leakage is the fraction of predictions in the wrong task's class set. All runs: 5 epochs/task, lr 10^{-3}, B=8, rank 4, seeds 0–4.

Arms. R1 per-task specialists (oracle upper bound); R2 joint offline training; A1 sequential full fine-tuning (SeqFT); A2 LoRA + boundary merge only; A3 A2 + replay; A4 A3 + dual-level distillation; A5 the full method (A4 + utility hinge); L2 a single continual adapter never reset. A2–A5 share the same LoRA parameterization, task order, and within-task objective, differing only in the enabled preservation components; A1 is a full-parameter sequential baseline, and R1/R2 are specialist and offline references.

Metrics. Final class-IL accuracy (A_{Final}), final task-IL accuracy, leakage; Forgetting $F = \frac{1}{T-1}\sum_{j<T}[\max_{s \geq j} a_{s,j} - a_{T,j}]$ and Plasticity $\frac{1}{T}\sum_j a_{j,j}$ from the full accuracy matrix; and two selection-level metrics computed on old-task slides: *Selection Jaccard* between the evidence set chosen right after the task was learned and the set chosen at the end of the sequence, and *utility retention* $\Delta U = U_{\text{final}} - U_{\text{own}}$ (values reported in the text).

Statistical discipline. With n=5 seeds we report mean±sd and *win counts* over seed-paired comparisons; we call a difference *systematic* only at 5/5 and *directional* at 4/5, and we do not report p-values at this seed count. All gates below were registered before the corresponding runs.

5. MAIN RESULTS

Accuracy. The full method gains +34.6 class-IL points over SeqFT (0.8239 vs. 0.4774; 5/5 seeds), +10.6 task-IL points (5/5), and cuts leakage by 34.0 points (5/5), closing most of the gap to the per-task oracle and surpassing joint training's final accuracy. A2 shows no systematic difference from A1 in the paired five-seed comparison (3/5)—the substrate result anticipated in Sec. 3.3. 3.3.

Forgetting is halved, plasticity is untouched. A5's class-IL forgetting is 0.0539 vs. 0.1102 for replay alone; its task-IL forgetting is 0.0080. Plasticity remains within a narrow 0.85–0.88 range across arms; the large retention gains are not accompanied by an obvious collapse in new-task learning.

Forgetting where to look, measured directly. Selection Jaccard separates the field by two orders of magnitude: unprotected arms re-select almost nothing of their original evidence (0.0023, 0.0010), replay recovers some (0.0864), and distillation-based arms retain the most (0.1752, 0.1613). Exact evidence-set overlap and diagnostic performance need not move monotonically together: A4

Table 1. Continual-learning main table (mean±sd over 5 seeds; class-IL unless noted; flat selector substrate—the canonical main-table configuration, with hierarchical-policy contrasts reported in the text). Rows form a lifecycle progression: full fine-tuning → PEFT substrate → memory → behavioral preservation. Dashes: not applicable. Leakage: lower is better.

Method	$A_{\text{Final}}\uparrow$	Forgetting↓	Plasticity↑	Sel. Jaccard↑	task-IL↑	Leakage↓
Per-task specialist (R1)	0.8777±.020	—	0.8777	—	0.9027±.019	—
Joint (R2)	0.7789±.028	—	0.7789	—	0.8498±.029	—
SeqFT (A1)	0.4774±.103	0.5193	0.8669	0.0023	0.8086±.061	0.4408
Seq-LoRA, one adapter (L2)	0.7885±.015	0.0982	0.8555	0.0661	0.8941±.018	0.1254
LoRA-Merge (A2)	0.4466±.109	0.5798	0.8814	0.0010	0.8105±.036	0.4756
+ Replay (A3)	0.7778±.015	0.1102	0.8514	0.0864	0.9073±.018	—
+ Distillation (A4)	0.7972±.046	0.1063	0.8689	0.1752	0.8969±.019	—
PathSelect (A5)	**0.8239±.028**	**0.0539**	0.8515	0.1613	**0.9147±.014**	**0.1005**

retains slightly higher Jaccard, whereas A5 achieves better final accuracy and forgetting; we report both. The revised utility-retention metric orders arms consistently with accuracy: ΔU is -16.8 for the full method and $-21.2/-31.9$ for replay/distillation, versus $-220/-301$ for unprotected A1/A2 (L2: -27.5; warm-start W1 best overall at -12.8)—every arm loses old-task utility, but protected arms lose an order of magnitude less. We also note honestly that the A5$-$A4 margin flips sign under some task orders; the class-IL gain of the utility hinge is order-sensitive and we present it as such.

Hierarchy and dual-level distillation. Under the flat substrate the A5$-$A3 task-IL margin is $+0.74\pm1.93$ (3/5); the hierarchical policy strengthens the same contrast to $+3.28\pm2.40$ task-IL and $+5.76\pm3.42$ class-IL (both 5/5), while its *absolute* A5 accuracy stays within noise of flat ($-0.58/-1.20$, 1/5) with somewhat higher leakage ($+2.15$, 0/5)—the architecture was adopted under pre-registered criteria for amplifying and stabilizing the method contrast, and we report both sides. Removing group-level distillation under the hierarchy costs 3.95 task-IL and 3.71 class-IL points (both 5/5), supporting an additional contribution from group-level preservation on top of the patch-level pathway.

Preservation design can outweigh buffer size. In this controlled comparison on the hierarchical configuration, the full method with a 128-entry memory outperforms replay-only with 512 entries ($+3.06$ task-IL, 5/5), indicating that *what* is stored—behavioral snapshots consumed by distillation and the hinge—can matter more than how much.

6. UPDATE-PROTOCOL ABLATIONS (PRE-REGISTERED)

All arms below were registered with two-sided decision rules before any run; win counts over 5 seeds decide, and the incumbent is retained on a split. One pre-registered expectation was refuted; we report it as measured.

Warm start (W1). Task 1 trains the full base; LoRA thereafter. Class-IL shows no systematic difference (W1 -1.80pp mean, better in 2/5 seeds; task-IL -0.96, 1/5), so the parameter-efficient incumbent is retained under the pre-registered split rule. Warm-starting does reshape selection: Jaccard rises to 0.245 ($+0.06$ over A5, 4/5) and ΔU is the best of any arm (-12.8)—changing *what* is selected without changing outcomes, echoing the removed-components pattern. Base quality matters precisely when nothing else protects: without preservation losses, W1B improves over A2 by $+2.86$ class-IL (4/5) with -1.59 leakage (4/5); with them, the advantage dissolves—preservation subsumes initialization.

One adapter vs. fresh-per-task (L2 vs. A5). Skipping the boundary reset caps the total drift at rank 4; fresh adapters allow rank up to 16. The capacity matters: the single-adapter variant loses 3.54 class-IL (A5 better in 4/5) and 2.06 task-IL (4/5) points, empirically supporting the fresh-adapter design the merge protocol assumes. (Bare, L2B edges A2 by $+1.37$ task-IL, 4/5—mild continuity benefits exist, but capacity dominates once preservation is active.)

Post-hoc composition (C1 sum, C2 average). The pre-registered expectation was refuted—informatively. Residuals trained independently from a shared θ_0 and then summed beat the bare sequential substrate on every metric in every seed ($+24.6$ class-IL, $+6.6$ task-IL, -24.9 leakage; all 5/5): without preservation, sequential updating is actively destructive—each task's update freely overwrites its predecessors—whereas independent residuals never interfere during training and compose well, echoing the task-arithmetic literature [12]. Composition is not the endpoint: its defining independence excludes replay, distillation targets require an accumulated model, and the full method beats it decisively ($+13.2$ class-IL and -12.6 leakage, both 5/5; $+3.8$ task-IL, 4/5). Sum and average are interchangeable here (no systematic difference). The refined claim: *sequential consolidation earns its place only in combination with behavioral preservation—the component that requires sequence to define*. Composition also attains the highest selection Jaccard of any arm (0.275) while trailing by 13 accuracy points—a second instance, after warm start, of interventions that stabilize *what is selected* without improving *what is diagnosed*.

Damped merging (A5H, α=0.5). Halving each fold-in leaves class-IL within noise (2/5); full merging is directionally better on task-IL ($+0.98$, 4/5) and leakage (-1.15, 4/5), so α=1 is retained. Damping raises Jaccard (0.238 vs. 0.161)—the third such instance: selection stabilizes, accuracy does not.

Replay budget. Sensitivity to the replay draw and to memory-policy variants is deferred to the extended version.

7. REMOVED COMPONENTS

Three architectural components failed their pre-registered gates and were removed from the method; we report them because removal is not hiding. In the general parameterization the selectors accept $[\text{feat}; q_\tau; e; \tilde{b}] \in \mathbb{R}^{1537}$; the main configuration zero-fills the conditioning blocks after the corresponding variants failed their gates. The task-conditioning query q_τ failed its gate; separately, task identity is already highly decodable from slide-average features (98.21%) and group prototypes (98.57%), consistent with—though not proving—the hypothesis that explicit conditioning adds little new information on this benchmark. It nonetheless carries a reproducible secondary effect, reducing leakage by 5.92 points (5/5), which we retain as an observation, not a mechanism. Group-level semantic priors and stateful (sequential) selection likewise failed their gates; the group-level prior carries its own secondary finding, reducing quota-distribution divergence by 0.005 (-0.005 ± 0.004, 5/5) without translating into accuracy. Full gate cards appear in the extended version.

8. LIMITATIONS AND DISCUSSION

Five seeds and win-count discipline bound what we can claim; we report no p-values by design. Task boundaries are known during training (organ cohorts arrive staged), which is clinically motivated but weaker than task-free online CL; our inference, however, is fully task-free. We compare within parameter-isolation, rehearsal, and functional-regularization families; parameter-importance methods (e.g., EWC) are complementary and deferred. Results use one foundation model (CONCH) and one budget ($B{=}8$); the removed-component analysis suggests our benchmark's tasks are unusually separable, and harder mixtures may re-open the case for explicit task conditioning.

9. CONCLUSION

Freezing everything except a budgeted selection policy turns continual learning for WSIs into a question that can be asked cleanly and answered mechanistically: models do forget where to look, merging alone does not stop them, and training-time behavioral preservation—replay, dual-level selection distillation, and a utility hinge over stored snapshots—substantially mitigates it, at zero inference-time cost.

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